

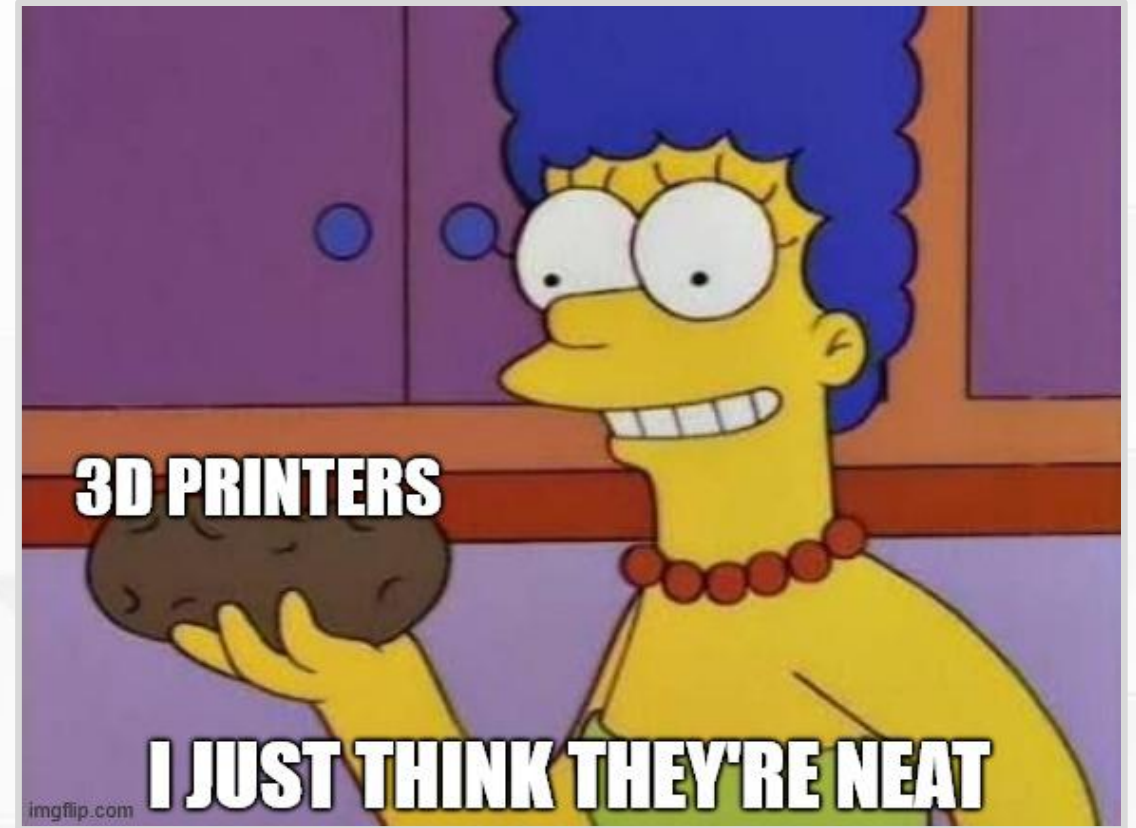


FFF Printing Fundamentals

Part 4: Advantages

An Amazing Tool

- Playground for **creative design** and **creative problem solving**.
- Massive utility for cheap.
- Diverse applications.
 - Art projects, toys, DIY and repair, tools for work and research, etc.
- Bring your ideas to life.
- Fast turnaround time.
- Relatively low barrier to entry.

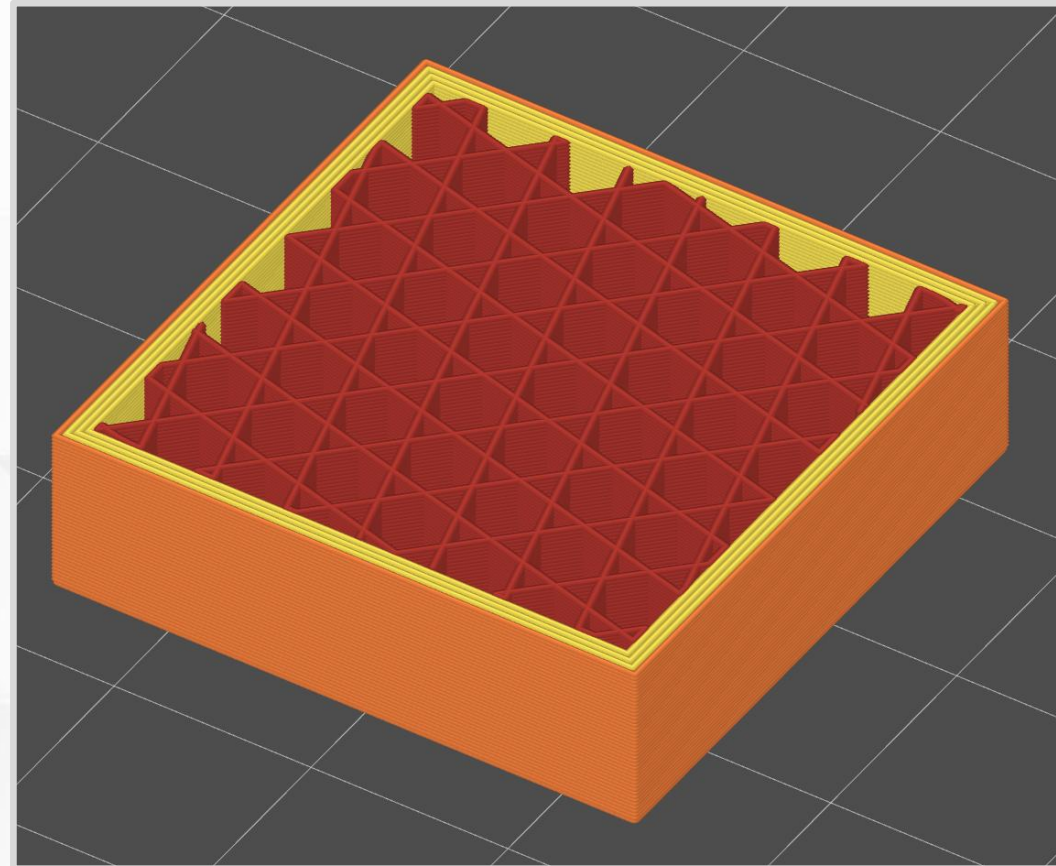


An Amazing Community

- RepRap festivals, maker fairs, local clubs and maker spaces, ...
- Projects that make a difference. Examples include:
Toddler mobility trainer by MakeGood
- People share cool projects and ideas. Examples include:
Full spectrum printing
BumpMesh.com
Voron
Countless free, downloadable models
- YouTube makers with educational content. Examples include:
CNC Kitchen
Maker's Muse
JanTec Engineering
3D Printing Nerd
Teaching Tech (retired)

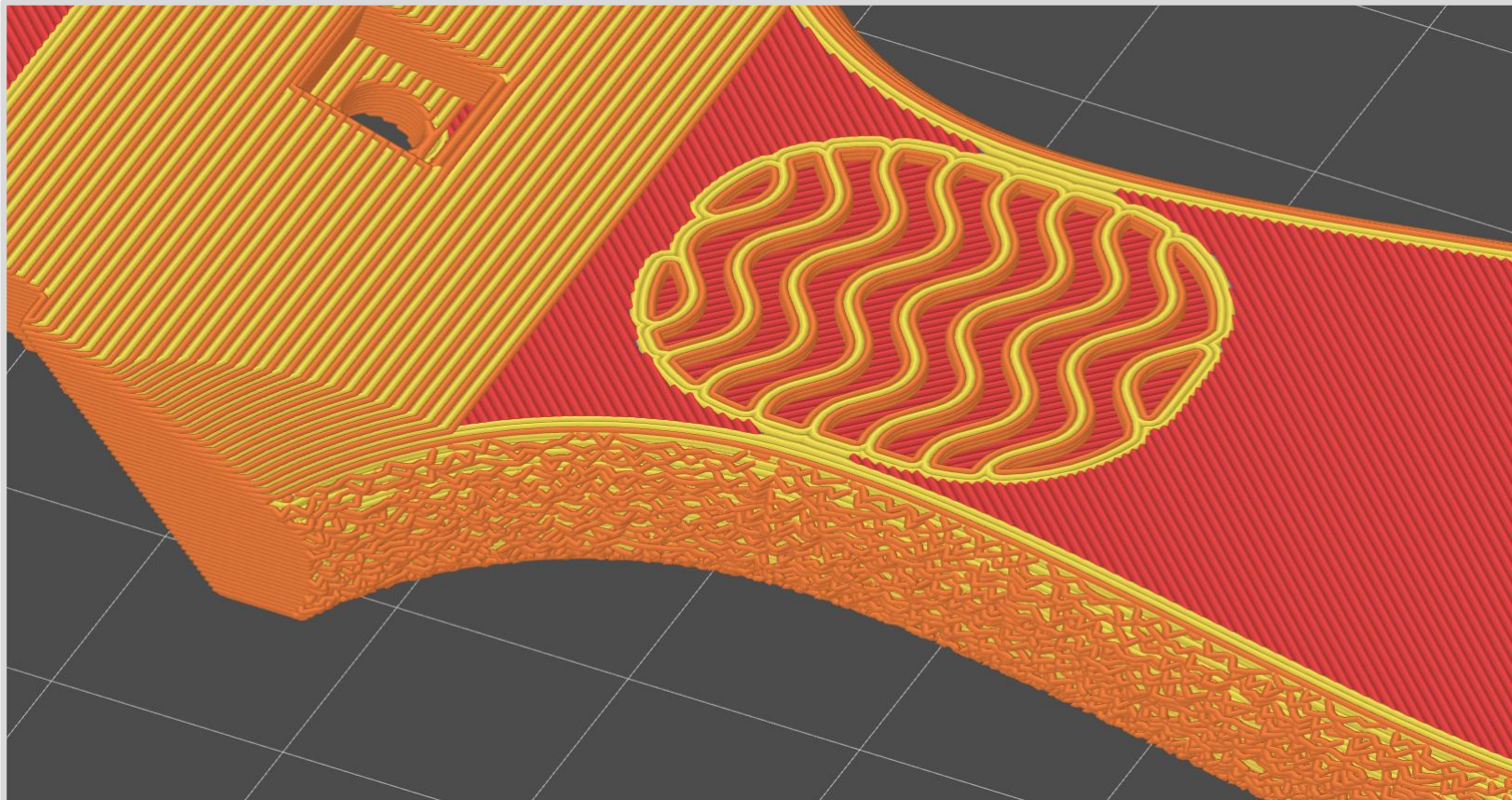
FFF Advantages

- Make lots of shapes with tiny “building blocks.”
 - Typical extrusions are **0.2 mm tall** (1-2 pieces of paper) and **0.4 mm wide**.
- Not restricted to uniform, solid objects.
 - Easily create internal geometry.
 - Great at making hollow objects.



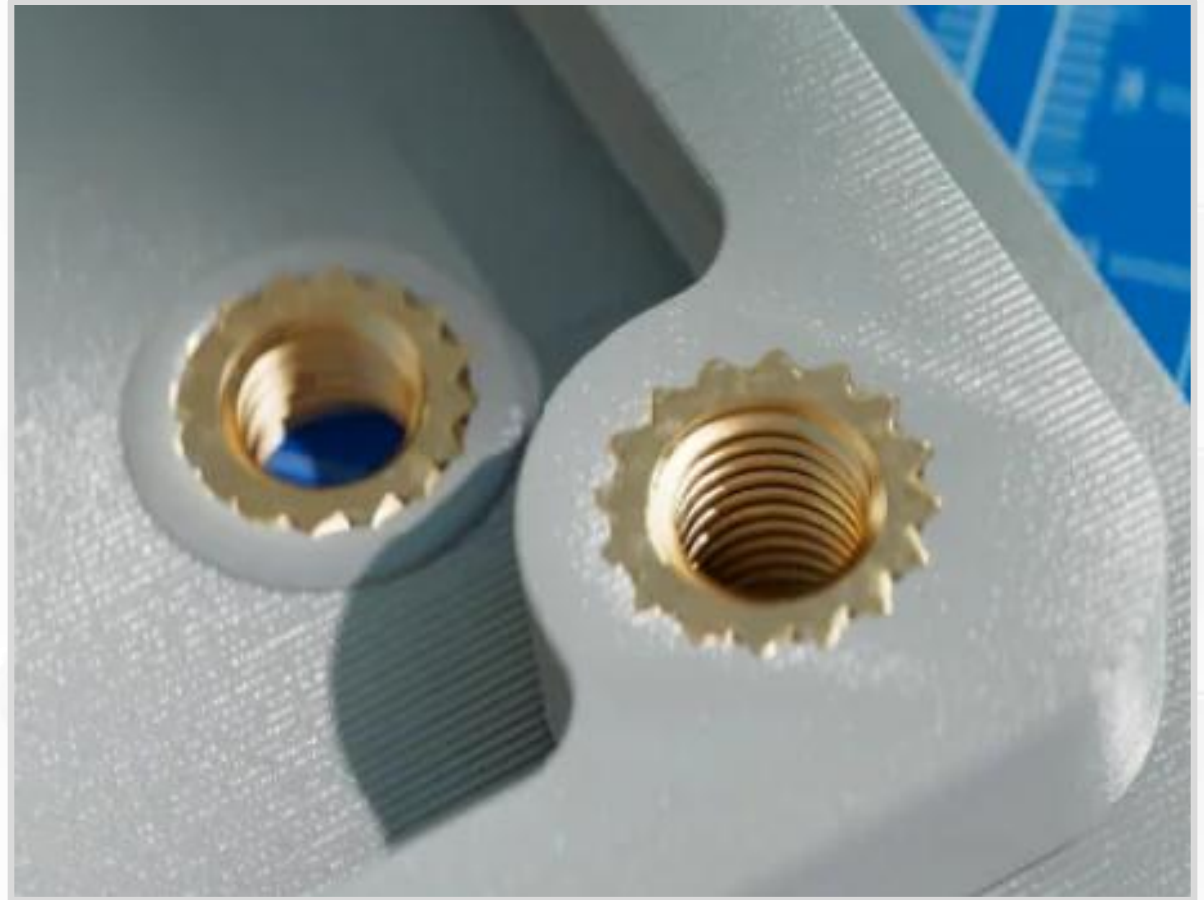
Custom Slicing

- Change slicer settings for specific regions or surfaces. Too many to list.
- *Example:* Add texture to handles with fuzzy skin (bumpy extrusions).



Heat Set Inserts

- Threaded anchor for screws. Why?
- Plastic is soft → Screws easily wear out plastic holes.
- Install with a hot soldering iron.
- Locks in place when it cools.
- Very durable.



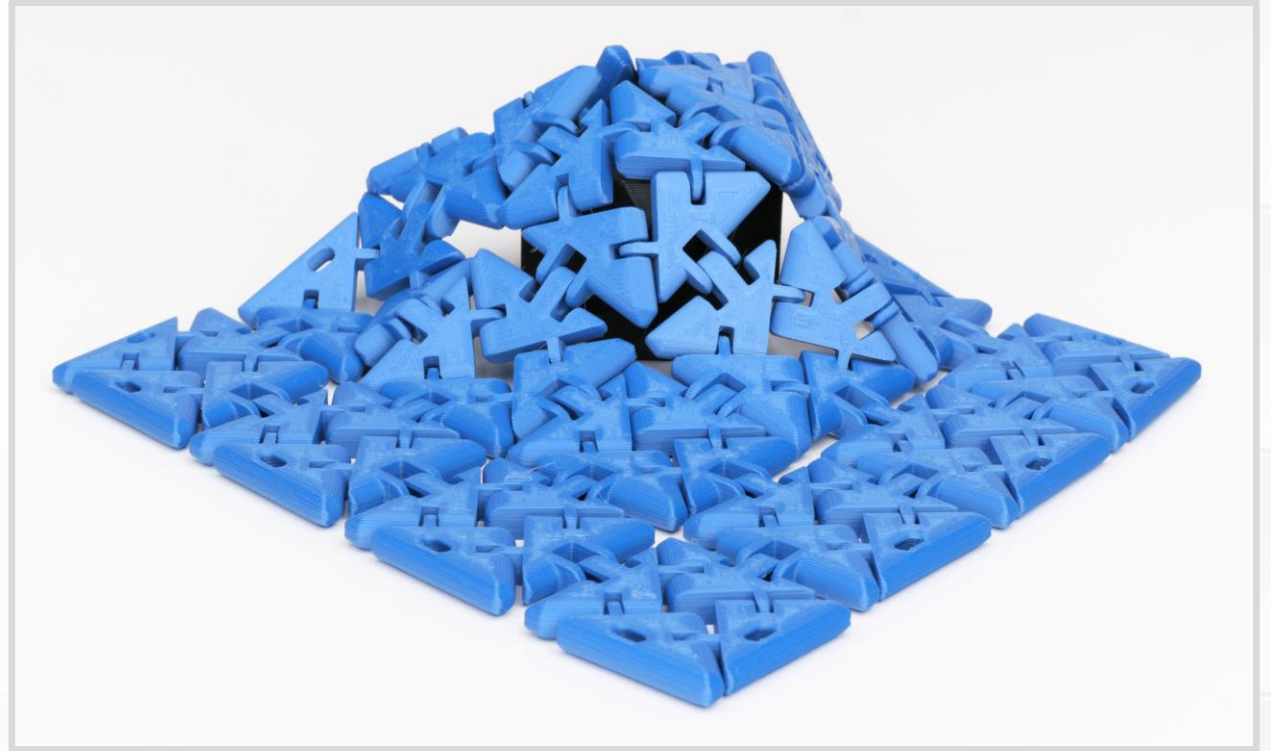
(Pictures from [CNC Kitchen Store](#))

Captive Parts

- Insert an object into a pocket or void in the middle of a print.
 - Nuts and fasteners.
 - Magnets.
 - Weights.
 - Reinforcing bar or plate.
- Locked inside when the print finishes.
- Object **must** sit below the current layer, or it creates a collision with the toolhead.
- A little inconvenient.
 - Need to be at the printer at the right layer.

Print-in-Place (PIP)

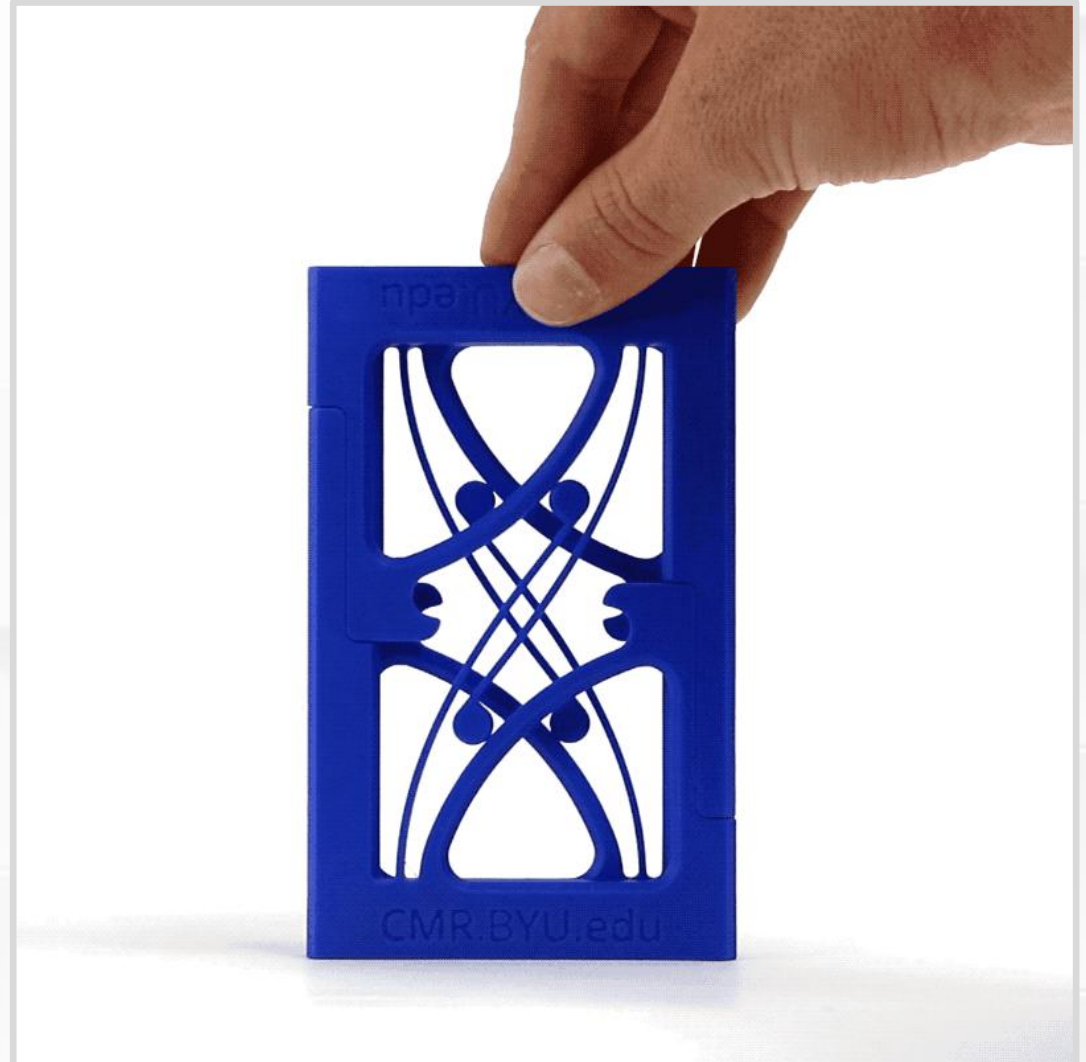
- Single print with interlocking parts.
- Assembled when the print finishes.
- *Example:* Hinges, chain links, chainmail.
- Flexify collection by FlyingGyroscope



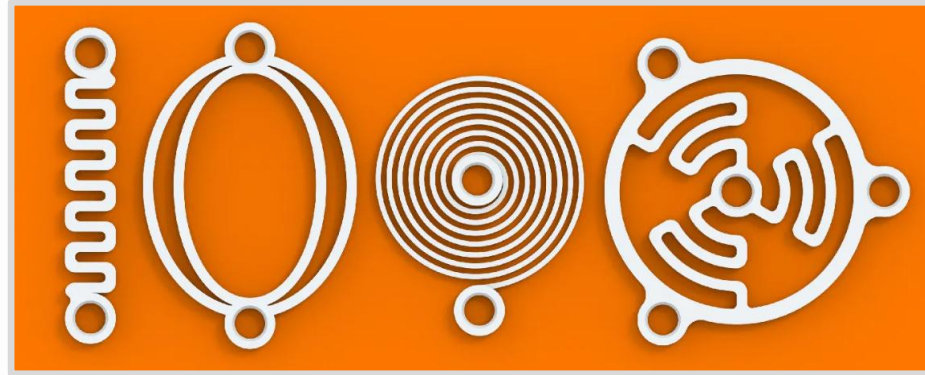
Functional Flexing

- Springs and **compliant mechanisms**.
- Features and shapes that function by bending and flexing.
- Elastic, springy materials.
 - No permanent deformation.
- Remember part strength and layer lines.

(Animation, Constant Force
Mechanism by BYU CMR)



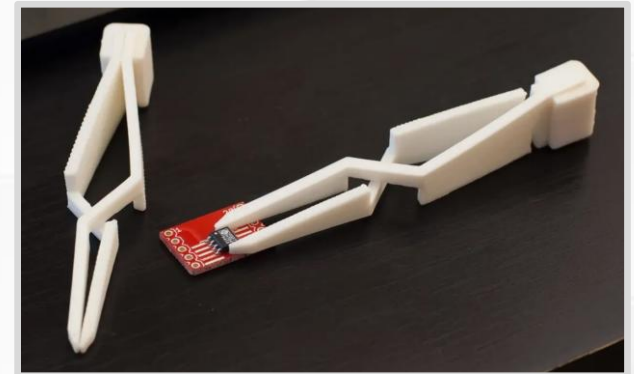
Springs by VC Design



Clip by AE



Cross tweezer by joo



Living hinge by DrJones

